# Local AI Orchestrator

System Guide and Architecture

Private, local-first executive work orchestration on macOS
v1 scaffold — clarification implemented, execution graph in progress

Generated: 2026-05-20

# Table of contents

# 1. What this system is

A private, local-first AI work orchestrator running entirely on your Mac. It is **not a chatbot**. It is a system that takes vague work requests ("prepare me for tomorrow," "summarize the board notes"), turns them into structured *work packets*, asks you strong clarifying questions until the request is concrete and safe to execute, scores readiness on nine dimensions, and — eventually — executes the well-scoped work overnight using local Ollama models.

The design intent is two-fold:

- **Clarify before doing anything heavy.** Vague input never produces low-quality output; it produces questions.
- **Keep everything on your machine.** No cloud calls unless every one of six policy gates passes. No external sends in v1. No service exposure beyond localhost.

## The everyday claim

*If you give it a vague request during the day, it gives you back a list of questions. If you answer those questions, it scores you ready. Overnight it produces a morning brief that respects what you said. Nothing leaves your machine.*

# 2. What this system is NOT

The negative space matters as much as the positive space. The orchestrator deliberately doesn't try to be these things:

- Not a chatbot — for conversations with a local model, use Open WebUI at http://localhost:3000.
- Not multi-user — single Mac, single operator, no auth.
- Not for sending anything externally — email, Slack, calendar invites, GitHub PRs, production API writes are blocked at the safety layer in v1.
- Not auto-cloud — Claude requires six gates; the day-unlock switch does not open them.
- Not running models in the background all day — DAY_MODE blocks every local-* model group unless you deliberately unlock.
- Not exposed beyond localhost — every port binds to 127.0.0.1.
- Not a knowledge base by itself — it writes to your Obsidian vault and reads from ~/LocalAI/inbox/; both are yours.
- Not auto-armed at install — the nightly launchd job is generated but never loaded without your explicit terminal command.

# 3. Mental model and the clarify-then-execute loop

The system is built around one loop, repeated as often as you have work. Read this table once and the rest of the architecture will follow:

| Stage | What happens | Who acts |
|---|---|---|
| 1. Capture | You give it a title and a vague description. | You |
| 2. Clarify | The system builds an initial work packet, scores readiness on 9 dimensions, and generates up to 7 strong quest | System |
| 3. Answer | You respond in a markdown file or via the panel. | You |
| 4. Re-score | The system parses your answers, rescores, and either marks the packet READY_FOR_OVERNIGHT (>= 0.85) | System |
| 5. Execute | Overnight, in NIGHT_MODE, the execution graph reads sources, calls local models (through the safety gate, eva | System (scheduled) |
| 6. Review | You read the output in ~/LocalAI/output/<date>/ and the Obsidian vault; approve memory candidates for the long | You |

Two key properties of the loop:

- **The system never executes without your consent.** Readiness threshold is the gate; below 0.85 (or 0.90 for high-stakes) the work refuses to start.
- **The clarification step is valuable by itself.** Most weekly value comes from being forced to be explicit about quality, audience, sources, and escalation — even when overnight execution never runs.

# 4. Readiness scoring in depth

Readiness is a deterministic weighted sum over nine binary dimensions. Each dimension is 1 if present in the packet, 0 if missing. The score lands in [0, 1].

| Dimension | Weight | What it captures | Blocking? |
|---|---|---|---|
| objective | 0.18 | Is the desired outcome explicit? | yes |
| output_format | 0.14 | Is the output shape specified? | yes |
| sources | 0.14 | Which folders/files are in scope? | yes |
| privacy_cloud_policy | 0.14 | Is cloud explicitly allowed or forbidden? | yes |
| quality_threshold | 0.12 | What quality bar to optimize for? | no |
| audience | 0.10 | Who is the output for? | no |
| assumption_policy | 0.08 | May the system infer when uncertain? | no |
| escalation_policy | 0.06 | When should it stop and ask? | no |
| success_criteria | 0.04 | What makes the work done? | no |

## Thresholds and status mapping

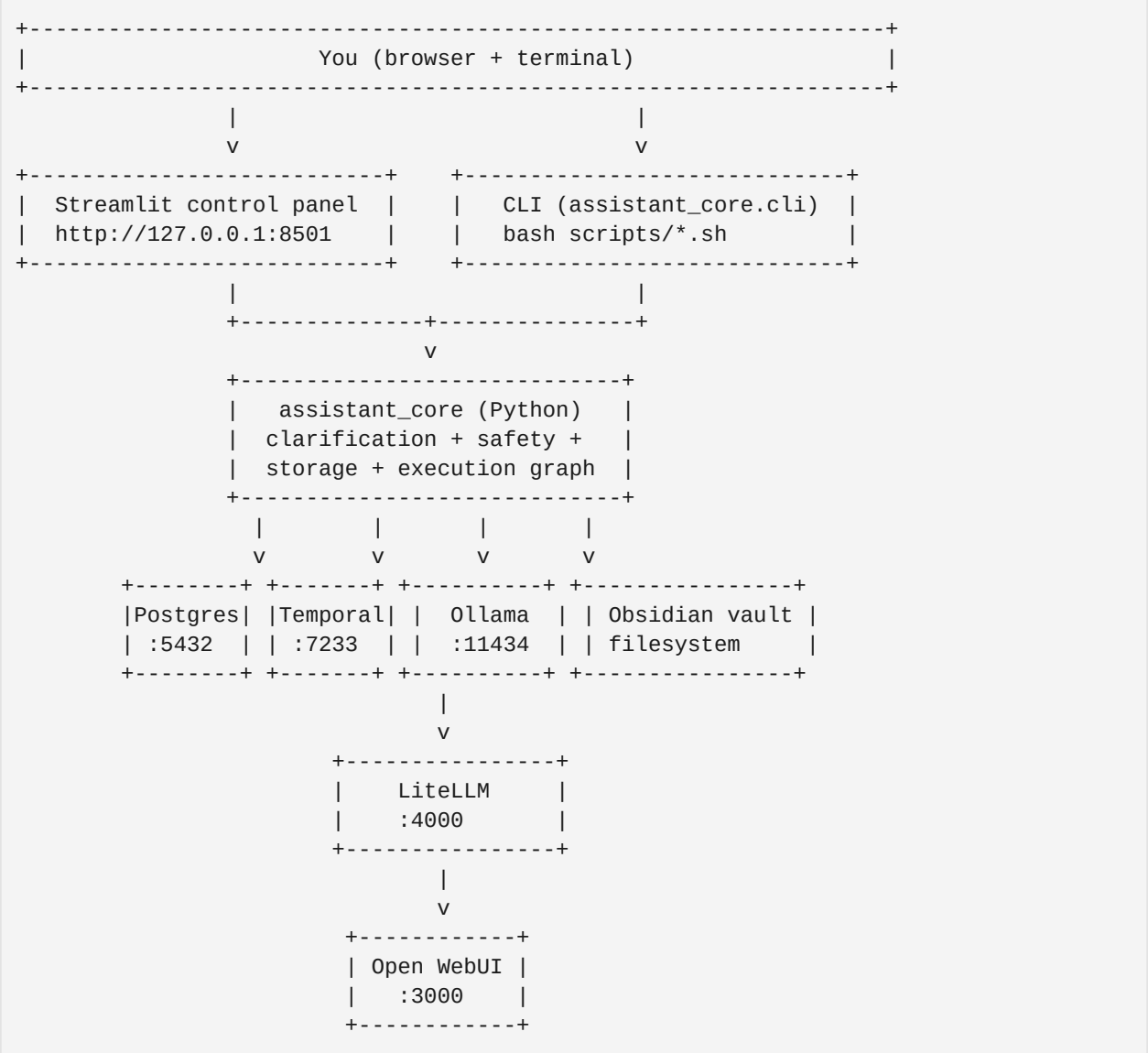
| Score range | Status | Eligible for overnight? |
|---|---|---|
| ≥ 0.90 | READY_HIGH_CONFIDENCE | Yes (required for high-stakes) |
| 0.85 to 0.89 | READY_FOR_OVERNIGHT | Yes for normal packets; no for high-stakes |
| 0.65 to 0.84 | NEEDS_CLARIFICATION | No; another question round |
| < 0.65 | UNDERDEFINED | No; major gaps remain |

**High-stakes auto-detection**: the work packet builder scans title and description for any of: board, CEO, legal, investor, high-stakes. If found, high_stakes=True is set and the threshold becomes 0.90.

**Blocking vs non-blocking gaps**: the four blocking dimensions (objective, output_format, sources, privacy_cloud_policy) are weighted heavily and must all be 1 to even reach 0.85. The other five contribute to fine-tuning but cannot individually keep a packet from going ready.

# 5. Architecture at a glance

Nine components, all running locally, talking to each other on 127.0.0.1.




Solid lines: required. LiteLLM and Open WebUI are optional in v1 — the panel talks to Ollama directly through assistant_core.llm.ollama_admin.

## What runs where

| Component | Lifecycle | Started by |
|---|---|---|
| Streamlit panel | Host Python process | bash scripts/start_control_panel.sh |
| assistant_core CLI | Host Python process (one-shot) | bash scripts/*.sh |
| Postgres | Docker container | bash scripts/start_services.sh |
| Temporal + Temporal UI | Docker containers | bash scripts/start_services.sh |

| Component | Lifecycle | Started by |
|---|---|---|
| Open WebUI | Docker container | bash scripts/start_services.sh |
| Ollama | Host service (brew) | brew services start ollama |
| LiteLLM | Host Python process | bash scripts/start_litellm.sh |
| Temporal worker | Host Python process (long-running) | bash scripts/start_temporal_worker.sh |

**Important consequence**: bash scripts/stop_services.sh only stops the Docker containers (Postgres, Temporal, Temporal UI, Open WebUI). It does not touch Streamlit, Ollama, or LiteLLM — those are host processes and must be stopped separately (Ctrl+C their terminals, or kill $(lsof -nP -iTCP:<port> -sTCP:LISTEN -t)).

# 6. Components: what each does and why

## Python orchestrator (assistant_core)

The brain. Pure Python, no framework.

### What it does

Holds the deterministic clarification logic, the safety gates, the work-packet schema, the storage layer, and the (scaffolded) execution graph. Has no opinion about UI — both the Streamlit panel and the CLI call into it. Tested in isolation via pytest.

### Why this tool

Pure Python keeps the core auditable, easy to reason about, easy to test. Pydantic for schemas (validation at type boundaries). YAML for human-editable config. psycopg for direct Postgres I/O without an ORM layer that hides intent.

### Where to look

```
assistant_core/{clarification,llm,memory,execution,temporal_app}/
```

## Streamlit control panel

Local web UI on :8501.

### What it does

Renders the status strip (mode, day unlock, services up/down), seven tabs (Dashboard, Create Work Packet, Clarification, Models, Schedule, Execution, Artifacts), and is the only UI surface for safe model load/unload.

### Why this tool

Streamlit lets us iterate in pure Python without writing a frontend. Tabs are stateless re-renders — safe by construction. Editing config or invoking Ollama always goes through assistant_core, never bypasses the gate.

### Where to look

```
app/control_panel.py
```

## PostgreSQL

Operational state store.

### What it does

Holds work_packets, clarification_questions, execution_runs, model_calls, evaluations, artifacts, memory_candidates. Migrated automatically on first container start (db/migrations/001_init.sql is mounted into Postgres' init dir).

### Why this tool

Postgres because we want durable, queryable state with relational integrity across the seven tables — work packets relate to questions, runs, model calls, and evaluations. JSON columns for cloud/source/output policy let us evolve without migrations.

### Where to look

```
db/migrations/001_init.sql + assistant_core/storage/
```

## Temporal

Durable workflow engine on :7233 (UI on :8233).

### What it does

Manages overnight workflows: clarification, execution, manual resume. Survives process crashes; can pause and resume long-running work. The worker process (scripts/start_temporal_worker.sh) executes activities defined in assistant_core/temporal_app/.

### Why this tool

Overnight work runs for hours, may hit transient model failures, and needs to be resumable. Temporal handles retries, timeouts, idempotency, and audit trail out of the box. Cron + bash would be fragile at this scale.

### Where to look

```
assistant_core/temporal_app/{workflows,activities,worker,client}.py
```

## Ollama

Local model runtime on :11434.

### What it does

Pulls models to ~/.ollama/models on disk, loads them into Metal VRAM on demand, serves inference via HTTP. The orchestrator never bypasses Ollama for local model calls — it always goes through this daemon.

### Why this tool

Ollama is the de facto local-model runtime on macOS Metal. Native Apple Silicon support, good keep-alive semantics, simple HTTP API, library of pre-quantized models. Lighter and easier to operate than running llama.cpp directly.

### Where to look

```
config/models.yaml maps groups -> Ollama tags
```

## LiteLLM

Model router and gateway on :4000.

### What it does

Translates a model-group name (local-main, local-coder, etc.) into the right Ollama tag or cloud endpoint. Configured with local-only fallbacks; cloud is intentionally not chained behind local.

### Why this tool

LiteLLM gives us a single OpenAI-style API for code that wants to swap models later, with fallback rules expressed declaratively. We keep the cloud route in the config but disable cloud at the safety layer so no blind escalation can happen.

### Where to look

```
config/litellm.yaml.example + assistant_core/llm/litellm_client.py
```

## Open WebUI

ChatGPT-style chat against local models, on :3000.

### What it does

A separate tool that lives next to the orchestrator. Talks directly to Ollama. Provides conversational access when you want it; the orchestrator deliberately does not offer chat.

### Why this tool

Sometimes you want to chat with a model. Open WebUI is a polished, mature tool for that. Co-installing it via docker-compose lets you have both surfaces.

### Where to look

```
docker-compose.yml — service: open-webui
```

## Obsidian vault

Human-readable long-term notes.

### What it does

The system writes outputs and memory candidates into ~/Obsidian/LocalAI-ChiefOfStaff/. Daily briefs, work packets, decisions, meeting notes, memory review queue. You read and curate these directly in Obsidian.

### Why this tool

Obsidian's plaintext markdown vault is durable, indexable, and yours. No lock-in. The orchestrator writes; you review. The vault becomes your long-term knowledge layer.

### Where to look

```
~/Obsidian/LocalAI-ChiefOfStaff/{01_Daily_Briefs, 02_Work_Packets, ...}
```

## Chroma (vector store)

Local semantic retrieval target.

### What it does

Scaffolded but not yet wired. Will hold embeddings of selected outputs and vault content for retrieval-augmented work.

### Why this tool

Chroma is simple, file-backed, no separate daemon. Right tradeoff for a single-user local system. Persistent client at ~/LocalAI/state/chroma.

### Where to look

```
assistant_core/memory/vector_store.py
```

## Docker Compose

Service orchestration for Postgres / Temporal / Open WebUI.

### What it does

One file, four services. Containers expose ports only on 127.0.0.1. Named volumes persist data across container recreates. The migration SQL is mounted into Postgres' init directory so the schema is created on first run.

### Why this tool

Docker keeps these three services consistent across machines without ceremony. They're stateful, multi-process, and have non-trivial init — containers handle that cleanly.

### Where to look

```
docker-compose.yml
```

# 7. The safety system in depth

The orchestrator is built around the assumption that you do not want the system to make decisions you wouldn't make yourself. Every model call, every cloud fallback, every external write passes through a deterministic gate in assistant_core/safety.py. Gates are written as *fail-closed* — an unknown tag, an empty env var, or a missing flag all default to **blocked**.

## The five gate functions

| Function | What it gates |
|---|---|
| assert_cloud_allowed | Six checks before any Claude call: cloud_fallback_enabled, work_packet_cloud_allowed, |
| assert_no_external_write | Always blocks in v1. Used at any future email/Slack/API boundary. |
| assert_original_file_write_allowed | Refuses writes outside the configured roots (output, archive, logs, work_packets, vault). |
| assert_heavy_execution_allowed | Refuses model execution in DAY_MODE unless day unlock is active or manual_override= |
| assert_model_allowed | Top-level entry point. Rejects cloud-claude-opus outright; then delegates local groups to a |

## Modes

| LOCALAI_MODE value | Meaning | Local model calls allowed? |
|---|---|---|
| DAY_MODE (default) | Workday: clarification + deterministic logic only. | No, unless day_unlock active |
| NIGHT_MODE | Overnight: heavy execution permitted. | Yes |
| MANUAL_RESUME | User-driven re-run of paused work. | Yes |

Mode is a single env var. The default is DAY_MODE. The overnight runner refuses to start unless NIGHT_MODE is set; the launchd plist sets it automatically when it fires.

# 8. The day-unlock switch

DAY_MODE blocks all four local-* model groups (local-main, local-secondary, local-coder, local-reasoner). To deliberately work with models during the day, the day-unlock switch flips a single boolean. Two channels:

- **Sentinel file** — ~/LocalAI/state/day_unlock.flag. Created by scripts/day_unlock.sh with a timestamp, hostname, and reason. Persists across terminal closes and reboots.
- **Environment variable** — LOCALAI_DAY_UNLOCK=true. Per-shell, ephemeral.

Either being on opens the gate. Cloud and external-write gates are **not** affected. The Streamlit panel renders a red banner with the flag contents while unlock is ACTIVE so the state is impossible to miss.

## Sentinel file format

```
$ cat ~/LocalAI/state/day_unlock.flag
unlocked_at: 2026-05-17T03:48:22Z
unlocked_by: user@host
reason: spot-checking the morning brief
```

Contents are informational only — the safety layer never parses them. An empty touch ~/LocalAI/state/day_unlock.flag also unlocks. The structure is for you (or for a future you reviewing why the flag was left on).

Full design rationale: docs/day_unlock.md. Locking again: bash scripts/day_lock.sh.

# 9. Auto-execution (launchd nightly job)

Default state: **OFF**. The orchestrator can be scheduled to run overnight via macOS launchd, but the plist is never armed automatically. Arming requires three deliberate steps in the terminal.

## The four states

| Project plist | Installed in LaunchAgents | Loaded in launchctl | State |
|---|---|---|---|
| no | no | no | fully off, nothing installed |
| yes | no | no | off (default after install_core) |
| yes | yes | no | off, installed but not loaded |
| yes | yes | yes | ARMED, will run nightly at night_mode_start |

## Arming (three deliberate steps)

```
# 1. Verify the configured time is what you want.
#    Edit it in the panel's Schedule tab or in config/assistant.yaml.

# 2. Regenerate the plist with that time:
bash scripts/create_launchd_plist.sh

# 3. Install into LaunchAgents and load:
LOCALAI_WRITE_LAUNCHD=true bash scripts/create_launchd_plist.sh
launchctl load ~/Library/LaunchAgents/com.localai.orchestrator.nightly.plist
```

## Disarming

```
launchctl unload ~/Library/LaunchAgents/com.localai.orchestrator.nightly.plist
# Optionally also delete the installed copy:
rm ~/Library/LaunchAgents/com.localai.orchestrator.nightly.plist
```

## How to verify the state from anywhere

```
launchctl list | grep com.localai.orchestrator.nightly
ls ~/Library/LaunchAgents/com.localai.orchestrator.nightly.plist
ls launchd/com.localai.orchestrator.nightly.plist
crontab -l | grep -i localai
```

Full design rationale: docs/auto_execution.md.

# 10. End-to-end data flow

From a vague request to overnight output, in nine steps. The next section walks the same flow with concrete JSON.

| Step | What happens | Storage |
|---|---|---|
| 1 | You: bash scripts/create_work_packet.sh "Title" "Description" | — |
| 2 | build_initial_work_packet builds a WorkPacket pydantic model. | in memory |
| 3 | score_readiness assigns 0/1 per dimension, computes weighted score. | in memory |
| 4 | generate_questions emits up to 7 ranked clarification questions. | in memory |
| 5 | create_work_packet + save_questions write the row and questions. | Postgres: work_packets, clarification_questions |
| 6 | You answer in a markdown file. bash scripts/answer_questions.sh parses. | — |
| 7 | update_packet_from_answers extracts fields. Rescore. If READY, mark status. | Postgres: update work_packets |
| 8 | Overnight runner (manual or via launchd at NIGHT_MODE) finds READY packets, submits | Postgres: packets; submits Temporal workflows. |
| 9 | Workflow body (scaffolded) reads sources, calls models, evaluates, writes outputs. | Filesystem + Postgres: artifacts, model_calls, evaluations |

> Step 9 is still scaffolded as of this writing. The workflow is wired and the Temporal worker accepts the submission, but the activity body returns a placeholder dict instead of running the 18-step pipeline. That is the next feature-work chunk.

# 11. Worked example: a clarification round

A complete walk through a real clarification. The same call you would make from the CLI, with the exact JSON the system returns at each step.

## Step 1: vague request

```
bash scripts/create_work_packet.sh \
   "Morning prep" \
   "Prepare me for tomorrow from my notes."
```

## Step 2: response — rejected for clarification

```
{
  "work_packet_id": "f37b8a92-...-4c11",
  "status": "UNDERDEFINED",
  "readiness_score": 0.18,
  "questions": [
    {
      "category": "objective",
      "question": "What exact outcome should this work produce, and what should be c
onsidered out of scope?",
      "priority": 100, "blocking": true
    },
    {
      "category": "outputs",
      "question": "What exact output files or formats do you want at the end?",
      "priority": 95, "blocking": true
    },
    {
      "category": "sources",
      "question": "Which folders or files are in scope, and are any sources explicit
ly off-limits?",
      "priority": 90, "blocking": true
    },
    {
      "category": "privacy/cloud",
      "question": "Is cloud fallback explicitly allowed for this packet, or must it
stay fully local?",
      "priority": 88, "blocking": true
    },
    ...
  ]
}
```

Score 0.18 is below 0.65, so status is UNDERDEFINED. Four blocking dimensions are missing (objective is too generic to satisfy the check; the rest have nothing). The system refuses to proceed.

## Step 3: answer markdown

```
# examples/sample_answers.md

Outputs: morning brief, today's priorities, risks and blockers, meeting prep.
```

```
Sources: only ~/LocalAI/inbox/notes and ~/LocalAI/inbox/docs.
Cloud: do not use cloud fallback.
Audience: private user only.
Assumptions: infer low-risk priorities but label every assumption.
Quality: optimize for factuality, criticality, and actionability.
Stop conditions: stop for missing sources, unclear ownership, or anything
  requiring external action.
Success: I can read it in the morning and know what matters first.
```

## Step 4: submit answers

```
bash scripts/answer_questions.sh \
   f37b8a92-...-4c11 \
   examples/sample_answers.md
```

## Step 5: response — ready

```
{
  "work_packet_id": "f37b8a92-...-4c11",
  "status": "READY_FOR_OVERNIGHT",
  "readiness_score": 0.9,
  "database": "updated",
  "remaining_questions": []
}
```

All four blocking dimensions are filled (objective is the request itself, outputs and sources extracted from the markdown, cloud policy explicit). The five non-blocking dimensions add up to 0.40 / 0.40 (audience, quality, assumptions, escalation, success all answered). Score lands at 0.90 = READY_HIGH_CONFIDENCE.

## What the packet looks like in Postgres now

```
SELECT id, status, readiness_score, high_stakes
FROM work_packets
WHERE id = 'f37b8a92-...-4c11';

             id              |        status         | readiness_score | high_stakes
-----------------------------+-----------------------+-----------------+------------
 f37b8a92-...-4c11           | READY_FOR_OVERNIGHT   | 0.90            | false
```

# 12. Worked example: walking the safety gates

A Python REPL session showing each gate in turn. Useful for understanding what the gate functions actually do without reading the code.

## DAY_MODE blocks local-main by default

```
$ .venv/bin/python
>>> from assistant_core.safety import assert_model_allowed, SafetyError
>>> try:
...     assert_model_allowed('local-main', 'DAY_MODE', manual_override=False)
... except SafetyError as e:
...     print('blocked:', e)
blocked: Local model execution is blocked in DAY_MODE without explicit
manual override.
```

## Manual override unblocks it

```
>>> assert_model_allowed('local-main', 'DAY_MODE', manual_override=True)
>>># returns None silently &mdash; call would proceed
```

## Day-unlock flag does the same thing without per-call override

```
$ bash scripts/day_unlock.sh "REPL demo"
>>> assert_model_allowed('local-main', 'DAY_MODE', manual_override=False)
>>># returns None silently
$ bash scripts/day_lock.sh
>>> assert_model_allowed('local-main', 'DAY_MODE', manual_override=False)
SafetyError: Local model execution is blocked in DAY_MODE without explicit
manual override.
```

## Cloud Claude is always blocked here

```
>>> assert_model_allowed('cloud-claude-opus', 'NIGHT_MODE', manual_override=True)
SafetyError: Claude must be authorized through cloud policy checks, not
model selection alone.
```

Cloud Claude has its own dedicated gate. Even with NIGHT_MODE + manual_override + day unlock all set, you cannot reach Claude through this function. The right path is assert_cloud_allowed(...) which checks six independent conditions.

## Cloud Claude with the proper gate (still mostly fails)

```
>>> from assistant_core.safety import assert_cloud_allowed
>>> assert_cloud_allowed(
...     file_path='~/LocalAI/inbox/cloud_review/brief.cloud.md',
...     cloud_fallback_enabled=False,   # config gate -> raises
...     work_packet_cloud_allowed=True,
...     high_stakes=True,
...     local_quality_gate_failed=True,
...     anthropic_api_key='sk-...',
...     daily_budget_remaining=True,
...     cloud_review_dir='~/LocalAI/inbox/cloud_review',
```

```
... )
SafetyError: Cloud fallback is disabled by configuration.
```

Six different gate conditions; flipping any one false raises. To actually reach Claude you need to satisfy all six simultaneously. That's deliberate.

# 13. Worked example: loading and unloading a model

The Models tab and the underlying API calls. Same effect from either entry point.

## From the panel

- Browse to http://127.0.0.1:8501 -> **Models** tab.
- Day unlock must be ACTIVE if you're in DAY_MODE — run bash scripts/day_unlock.sh "reason" first if needed.
- Click **Load** next to deepseek-r1:8b. Spinner. Success toast.
- Model now appears in *Currently loaded* with size, expires_at, and Unload button.
- Scroll to **Quick test prompt**, pick the model, type a prompt, click Send.
- Click **Unload**. Model disappears from *Currently loaded*.

## What the panel does behind the scenes (curl equivalents)

```
# Load with 30-minute keep-alive
curl -X POST http://localhost:11434/api/generate \
   -d '{"model":"deepseek-r1:8b","prompt":"","keep_alive":"30m"}'

# What is currently in VRAM
curl http://localhost:11434/api/ps

# Send a prompt
curl -X POST http://localhost:11434/api/generate \
   -d '{"model":"deepseek-r1:8b","prompt":"Say hi.","stream":false}'

# Unload (note the string '0s', not int 0 &mdash; some Ollama versions parse
# int 0 as "use default 5m" and silently re-pin the model)
curl -X POST http://localhost:11434/api/generate \
   -d '{"model":"deepseek-r1:8b","prompt":"","keep_alive":"0s"}'
```

## Equivalent CLI

```
ollama list                    # what's on disk
ollama ps                      # what's in VRAM
ollama run deepseek-r1:8b      # interactive chat (not what the orchestrator uses)
ollama stop deepseek-r1:8b     # immediate unload (sends keep_alive=0s)
```

## What the gate sees

```
# Inside ollama_admin.load_model('deepseek-r1:8b', mode='DAY_MODE'):
1. resolve_group_for_tag('deepseek-r1:8b') -> 'local-reasoner'
2. assert_model_group_allowed('local-reasoner', 'DAY_MODE')
     -> calls assert_model_allowed
       -> not cloud-claude-opus, so skip first check
       -> 'local-reasoner' in LOCAL_MODEL_GROUPS, delegate
```

```
         -> assert_heavy_execution_allowed('DAY_MODE')
           -> day_unlock_active() reads ~/LocalAI/state/day_unlock.flag
           -> if ACTIVE: pass; if not: raise SafetyError
3. If gate passes: POST /api/generate with keep_alive='30m'
```

# 14. Worked example: editing day/night times

Say you want the nightly job to start at 02:00 instead of 01:30, and treat 06:30 as the end of night. Two ways: edit the config file directly, or use the panel.

## Via the panel

- Browse to the panel -> **Schedule** tab.
- Edit night_mode_start to 02:00 and night_mode_end to 06:30.
- Click **Save to config/assistant.yaml**. Atomic write; validation runs first.
- If the plist is already installed in LaunchAgents, a yellow warning appears with the regenerate+reload commands.

## Via the terminal

```
# Edit config/assistant.yaml by hand or with sed; the panel and the safety
# layer pick it up automatically on next read.

# Then if the plist is installed:
LOCALAI_WRITE_LAUNCHD=true bash scripts/create_launchd_plist.sh
launchctl unload ~/Library/LaunchAgents/com.localai.orchestrator.nightly.plist
launchctl load ~/Library/LaunchAgents/com.localai.orchestrator.nightly.plist

# Verify the new time is baked into the installed plist:
grep -A3 StartCalendarInterval \
  ~/Library/LaunchAgents/com.localai.orchestrator.nightly.plist
```

## What does NOT change automatically

- LOCALAI_MODE stays at whatever the shell sets it to. Editing the times does not auto-flip mode.
- The launchd plist installed in ~/Library/LaunchAgents is not regenerated until you run the script with LOCALAI_WRITE_LAUNCHD=true.
- Already-running services don't restart.
- Already-running Temporal workflows continue with whatever mode they were started in.

# 15. Sample artifacts (what a morning brief looks like)

When the overnight pipeline is fully wired, each run produces a dated directory under ~/LocalAI/output/YYYY-MM-DD/ containing the following nine files. The shapes are committed; only the placeholder content rendered today will change once the LangGraph executor is implemented.

## Directory contents

| File | Purpose |
|---|---|
| 00_STATUS.json | Machine-readable summary of the run (packet id, status, timestamps). |
| 01_MORNING_BRIEF.md | Top-line synthesis: what matters today, in priority order. |
| 02_TODAY_PRIORITIES.md | Ordered list of priorities with rationale and source citations. |
| 03_ACTIONS_BY_PERSON.md | Grouped by stakeholder: what each owns and what you owe each. |
| 04_DECISIONS_NEEDED.md | Items requiring your call before they can move. |
| 05_RISKS_AND_BLOCKERS.md | Flagged risks, with criticality and recommended next action. |
| 06_DRAFT_MESSAGES.md | Drafts of messages you might want to send. Never sent automatically. |
| 07_MEETING_PREP.md | Per-meeting prep notes: who, agenda, what to push, what to listen for. |
| 08_CLOUD_REVIEW_CANDIDATES.md | Items the local model flagged as candidates for cloud review (only used if cloud policy per |
| 09_AUDIT_LOG.md | What ran, what models were called, what the evaluator said. |

## Sample 01_MORNING_BRIEF.md (mock content)

```
# Morning Brief &mdash; 2026-05-17

## Today's three things

1. **Decide Q3 staffing for the Scope1 project.** Three weeks of notes
   point at a hire vs reallocate question. Source: 04_DECISIONS_NEEDED.md.
2. **Reply to the legal followup on the data sharing agreement.** Draft
   in 06_DRAFT_MESSAGES.md ready for your review.
3. **30-min prep for the board sync at 14:00.** Agenda + likely Qs in
   07_MEETING_PREP.md.

## Risks surfaced

- Vendor X has not confirmed renewal terms (deadline Friday).
- The migration runbook from 2026-05-10 still has two unowned steps.

## Sources used

- ~/LocalAI/inbox/notes/2026-W20-recap.md
```

```
- ~/LocalAI/inbox/transcripts/2026-05-15-scope1-1on1.md
- ~/LocalAI/inbox/docs/2026-Q3-staffing-draft.md

## Confidence

Medium-high. Six assumptions labeled in 02_TODAY_PRIORITIES.md.
```

Memory candidates extracted from this run land in ~/Obsidian/LocalAI-ChiefOfStaff/00_Inbox/Memory_Review/ for your approval before they enter long-term memory.

# 16. Configuration recipes

Three schedule profiles you might pick from.

## Recipe: "strict night worker" (default)

```
day_mode_start:   '07:30'
night_mode_start: '01:30'
night_mode_end:   '07:00'
```

Day window is long (07:30 - 01:30 next day). Night execution starts late and ends well before you wake up. Good for the typical knowledge-worker schedule. Heavy models run with the laptop on charger overnight; you wake up to the brief.

## Recipe: "early bird"

```
day_mode_start:   '05:30'
night_mode_start: '23:00'
night_mode_end:   '05:00'
```

Night starts at 23:00, leaving six hours of model time before the day window opens at 05:30. Suits someone who's at their desk by 06:00 reading briefs.

## Recipe: "weekday office, weekend lab"

```
day_mode_start:   '08:00'
night_mode_start: '20:00'
night_mode_end:   '07:30'
```

Longer night window (11.5 hours) for someone who runs experiments or larger batch work overnight. The day window is shorter, so models stay safely untouched during working hours unless you explicitly day-unlock.

## Threshold tuning

```
# Stricter: require near-perfect packets to go ready
minimum_readiness_for_overnight:   0.92
minimum_readiness_for_high_stakes: 0.95

# Looser: accept more borderline packets (NOT recommended)
minimum_readiness_for_overnight:   0.75
minimum_readiness_for_high_stakes: 0.85
```

Lowering thresholds is rarely the right answer. Below 0.85, blocking gaps are almost guaranteed and the system produces lower-quality output that you'll have to redo. Stricter is fine; looser is a tax on tomorrow's you.

# 17. Model selection by hardware

Pick tags appropriate for your VRAM. The default config assumes a beefy M2 Max with 77+ GiB Metal VRAM. Smaller machines need smaller tags.

| Hardware | local-main | local-secondary | local-coder | local-reasoner |
|---|---|---|---|---|
| 96 GB unified (M2/M3 Max) | qwen3:30b-a3b | llama3.3:70b | qwen3-coder:30b | deepseek-r1:8b |
| 64 GB unified | qwen3:30b-a3b | (skip 70B) | qwen3-coder:30b | deepseek-r1:8b |
| 32 GB unified | qwen3:14b | qwen3:32b | qwen2.5-coder:14b | deepseek-r1:8b |
| 16 GB unified | qwen3:8b | qwen3:14b | qwen2.5-coder:7b | deepseek-r1:8b |
| Linux NVIDIA 24GB | qwen3:30b-a3b | qwen3:32b | qwen3-coder:30b | deepseek-r1:8b |

## When you change tags

Both config/models.yaml and config/litellm.yaml must reflect the same tag for each group. The Models tab in the panel will pull the name from models.yaml for the gate; LiteLLM will use the litellm.yaml entry for routing.

```
# config/models.yaml
local-main: qwen3:30b-a3b      # change this

# config/litellm.yaml
- model_name: local-main
  litellm_params:
    model: ollama/qwen3:30b-a3b  # and this, in lock-step
    api_base: http://localhost:11434
```

## Why MoE 30B-a3b is the workhorse default

Qwen3's 30B Mixture-of-Experts model with 3B active parameters gives 30B-class quality at roughly 8B-class inference speed. Excellent fit for batch overnight work on Apple Silicon: weights load quickly, per-token latency is low, instruction following is strong for clarification-style tasks.

# 18. What's implemented vs scaffolded

| Capability | State | Notes |
|---|---|---|
| Deterministic clarification flow | Implemented | Builder, 9-dim readiness, question gen. 100% test-covered. |
| Postgres storage layer | Implemented | 7 tables, schema migrated, CRUD for packets/questions/runs. |
| Safety gates (5 functions) | Implemented | Parameterized tests over every local-* group and mode. |
| Day-unlock switch | Implemented | File + env channels, visible banner, fail-closed. |
| Streamlit control panel | Implemented | Header strip, 7 tabs, gate-aware UI. |
| Models tab (load/unload/test) | Implemented | CLI workflow surfaced: Load -> Quick test -> Unload. |
| Schedule tab (time editing) | Implemented | Atomic config writes, validation, launchd-status mirror. |
| Ollama integration | Implemented | Pull, list, load, unload, generate. |
| Temporal worker + workflow defs | Skeleton | Workflows + activities defined; bodies return placeholders. |
| LiteLLM routing | Config only | Started via start_litellm.sh. Used by future call sites. |
| Launchd plist generation | Implemented | Reads night_mode_start; arming stays manual. |
| Overnight execution graph (18 steps) | Scaffolded | EXECUTION_STEPS list exists; activity body is stub. Next chun |
| Quality-gated local retry loop | Scaffolded | Deterministic evaluator fallback. Retry orchestration TBD. |
| Pause/resume UI | Scaffolded | Buttons disabled. DB statuses exist. |
| Semantic memory indexing | Scaffolded | Chroma client exists. No indexer yet. |
| Claude cloud-review path | Scaffolded | Gates implemented; no caller wired. |
| Drafted messages | Scaffolded | Output template exists; nothing fills it yet. |
| PDF documentation builder | Implemented | scripts/build_pdfs.py (thin entry) + scripts/pdf_builders/ package |

# 19. File system layout

## Project repository

```
local-ai-orchestrator/
  AGENTS.md              rules for human/AI editors
  ONBOARDING.md          practical usage guide
  README.md              setup + ports + safety summary
  app/control_panel.py   Streamlit UI
  assistant_core/        Python package
    clarification/       work packet builder, readiness, questions
    execution/           18-step pipeline (scaffolded)
    llm/                 ollama_admin, litellm_client, model_policy
    memory/              Obsidian + Chroma writers
    temporal_app/        worker, workflows, activities
    safety.py            5 gate functions
    schemas.py           pydantic models for the whole system
    storage/             Postgres CRUD package (by-entity submodules)
    config.py            yaml config loader
    config_writer.py     atomic config edits
    scheduler_status.py  launchd + service health
  config/
    assistant.yaml       day/night times, thresholds, paths
    models.yaml          group -> Ollama tag mapping
    memory.yaml          vector store settings
    litellm.yaml         LiteLLM router config (gitignored)
  db/migrations/         SQL applied on Postgres first boot
  docs/                  this PDF and friends, plus design docs
  examples/              sample answers, sample packet
  launchd/               generated plist (gitignored)
  scripts/               bash entrypoints for every operation
  tests/                 pytest, isolated from real state
  docker-compose.yml     postgres + temporal + temporal-ui + open-webui
```

## Data directories (outside the repo)

```
~/LocalAI/
  inbox/
    notes/         your input notes
    transcripts/   call/meeting transcripts
    docs/          documents
    csv/           tabular sources
    cloud_review/  files marked for cloud review (Claude gate path)
  output/<date>/   generated artifacts (briefs, priorities, etc.)
  archive/         processed inputs
  logs/            runtime logs (incl. launchd stdout/stderr)
  state/           operational state (incl. day_unlock.flag)
  work_packets/    packet-specific files

~/Obsidian/LocalAI-ChiefOfStaff/
  00_Inbox/Memory_Review/   memory candidates awaiting approval
  01_Daily_Briefs/          morning briefs
  02_Work_Packets/          packet notes
```

```
03_Projects/              project pages
04_Meetings/              meeting notes
05_Decisions/             logged decisions
06_Stakeholders/          people pages
07_Playbooks/             reusable procedures
08_Prompts/               curated prompts
09_System_Logs_Summaries/ run summaries
99_Archive/               historical
```

# 20. Database schema

Seven tables in the localai database, all created from db/migrations/001_init.sql:

| Table | Purpose |
|---|---|
| work_packets | One row per packet: title, objective, status, readiness, policies (jsonb), raw and structured reques |
| clarification_questions | Per-packet questions across rounds; each can be answered or unanswered. |
| execution_runs | One row per overnight invocation, with Temporal workflow id, mode, start/end timestamps. |
| model_calls | Audit row per local-model or cloud call: task type, group, actual tag, sizes, success/error. |
| evaluations | Quality / grounding / completeness / contradiction scores from the evaluator, with recommended |
| artifacts | File-level pointers to generated outputs and their Obsidian copies. |
| memory_candidates | Items extracted from outputs as candidates for long-term memory; require user approval. |

## Looking at the data directly

```
# Open a psql shell against the local Postgres container
docker exec -it localai-postgres psql -U localai -d localai

# In psql:
\dt                                   -- list tables
SELECT id, title, status, readiness_score, created_at
  FROM work_packets ORDER BY created_at DESC LIMIT 5;

SELECT round_number, category, question, answered
  FROM clarification_questions
  WHERE work_packet_id = 'YOUR-UUID'
  ORDER BY round_number, priority DESC;
```

All tables use UUID primary keys. JSONB columns on work_packets allow policy evolution without migrations. Foreign keys enforce relational integrity.

# 21. Code roadmap (file-by-file)

Where to look when you want to change something.

| When you want to… | Look here |
|---|---|
| change clarification question wording | assistant_core/clarification/question_generator.py |
| change readiness weights or thresholds | assistant_core/clarification/readiness.py + config/assistant.yaml |
| change which fields the work packet has | assistant_core/schemas.py |
| change what "high stakes" detects | assistant_core/clarification/work_packet_builder.py |
| add a safety gate or tighten an existing one | assistant_core/safety.py |
| add a new model group | assistant_core/safety.py (LOCAL_MODEL_GROUPS) + config/models.yaml + co |
| change the day-unlock UI behavior | app/control_panel.py (_render_header) + scripts/day_unlock.sh |
| wire the actual overnight execution pipeline | assistant_core/execution/graph.py + assistant_core/temporal_app/activities.py |
| add a Postgres column | db/migrations/ (add a new file) + assistant_core/storage/ |
| change LiteLLM routing | config/litellm.yaml + assistant_core/llm/litellm_client.py |
| add a new tab to the panel | app/control_panel.py (new _render_xxx_tab function + register) |
| add a new background service status | assistant_core/scheduler_status.py |
| change what files Ollama admin lists | assistant_core/llm/ollama_admin.py |
| add a new editable config field | assistant_core/config_writer.py (EDITABLE_FIELDS) + config/assistant.yaml |
| change the launchd schedule template | scripts/create_launchd_plist.sh |
| regenerate these PDFs | scripts/build_pdfs.py |

# 22. Anti-patterns and pitfalls

Mistakes that are easy to make. Each one is real (some are things I, your co-author, almost made you do at some point):

## Loading both 30B models simultaneously to "test things"

Two 30B-class models in VRAM is 60+ GB on M2 Max. The OS starts shuffling, the first inference takes 30+ seconds, and you'll blame the orchestrator. Unload one before testing the other.

## Lowering readiness thresholds to "get things moving"

Tempting when a packet just won't pass. Almost always wrong: the blocking gaps are blocking *because* the system can't safely produce useful output without them. Lower the bar and you get a confidently wrong morning brief.

## Leaving day_unlock on overnight

The flag persists across reboots. The Streamlit banner makes it visible, but the banner is only loud if you're *looking* at the panel. Make day_lock.sh part of your end-of-day habit.

## Pointing the orchestrator at ~/Documents

~/LocalAI/inbox/ is the source directory by convention. Pointing the system at broader user data invites accidents: a tag indexer might pick up personal docs, an evaluator might leak content into outputs you share. Stay in the sandbox.

## Setting ANTHROPIC_API_KEY "just in case"

Setting the key satisfies one of the six cloud gates. If you also set cloud_fallback_enabled: true in config and one of your packets happens to mark cloud allowed, a high-stakes failure path could escalate. Don't pre-arm a gun you don't plan to fire.

## Skipping Temporal because it's "just clarification"

True today — clarification is synchronous Python and doesn't touch Temporal. But the moment execution is wired, the worker needs to be running or workflows queue and never run. Bring up the worker as part of the start sequence so you don't get surprised at 06:00.

## Editing files in ~/LocalAI/output// expecting them to persist

Outputs are regenerated on each run; the dated directory is overwritten. Anything you want to keep, copy to your Obsidian vault. The vault is the durable layer.

## Running `stop_services.sh` and assuming everything is off

That script only stops Docker containers. Streamlit, Ollama, LiteLLM, and the Temporal worker are host processes. They survive. If you want everything down: stop_services.sh + Ctrl+C the Streamlit tab

+ brew services stop ollama + Ctrl+C the worker.

# 23. Glossary

| Term | Definition |
|---|---|
| Work packet | The orchestrator's unit of work. A structured representation of a request with objective, sources, |
| Readiness score | Weighted sum of nine binary dimension checks, in [0, 1]. Threshold 0.85 to be eligible for overnig |
| Clarification round | One pass of: build packet -> score -> generate questions -> wait for answers. |
| DAY_MODE | Mode in which all local-* model groups are blocked at the safety gate unless day_unlock is active |
| NIGHT_MODE | Mode in which local model calls are allowed. Required by the overnight runner. |
| MANUAL_RESUME | Third mode value. Allows the overnight runner to proceed when you're explicitly re-running pause |
| Day unlock | An explicit, visible override that lets local models load in DAY_MODE. Implemented as a flag file |
| Model group | A logical role (local-main, local-coder, etc.) that maps to a specific Ollama tag in config/models.ya |
| Keep-alive | Ollama's per-model VRAM retention timer. Our Load uses 30m; Quick test uses 5m; Unload sen |
| Cloud-review path | The Claude Opus route, gated by six policy checks. Files must be in ~/LocalAI/inbox/cloud_revie |
| Auto-execution | The launchd job that runs run_ready_overnight.sh at night_mode_start. Off by default; arming is |
| Manual override | A per-call boolean (manual_override=True) that bypasses the day gate. Used in tests and direct |
| Sentinel file | A small file whose existence (not contents) signals a state. day_unlock.flag is one; the launchd p |
| Fail-closed | Default-to-block when input is ambiguous, missing, or unknown. Opposite of fail-open. The right |
| Memory candidate | A statement extracted from a run that might be worth keeping long-term. Lands in the Obsidian m |
| MoE | Mixture of Experts. Models like qwen3:30b-a3b have 30B parameters but only ~3B active per tok |

End of System Guide. For practical usage recipes and day-to-day operation, see the companion Onboarding PDF.